

Highlights

A Multi-Strategy Local Search Memetic Algorithm for Multimodal Traveling Salesman Problem

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- Proposes a Multi-Strategy Local Search Memetic Algorithm for multimodal TSP
- Implements a two-stage evolutionary framework with exploration and exploitation stages
- Incorporates niching selection and diversity maintenance mechanisms
- Validates performance on MSTSP test suite and TSPLIB benchmark instances
- Demonstrates superiority over comparative algorithms in obtaining diverse high-quality solutions

A Multi-Strategy Local Search Memetic Algorithm for Multimodal Traveling Salesman Problem

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ABSTRACT

Multimodal problems are prevalent in real-world applications. For the Traveling Salesman Problem (TSP), multiple distinct shortest paths may exist. A decision maker (DM) can then select one for implementation based on specific requirements. However, research on multimodal problems within the field of discrete optimization remains limited. To effectively solve multimodal TSP (MMTSP), this paper proposes a Multi-Strategy Local Search Memetic Algorithm (MSLS-MA). The algorithm employs a two-stage evolutionary framework, partitioning the population evolution process into distinct exploration and exploitation stages. It utilizes a greedy algorithm to construct the population initialization mechanism. To enhance the distribution within the decision space, the algorithm incorporates an environmental selection method based on niching selection and implements a diversity-based population maintenance mechanism. To improve convergence performance, MSLS-MA constructs a local search operator leveraging incremental computation. Concurrently, a population restart strategy is incorporated and triggered upon stagnation in local optima to maintain population diversity. To evaluate the performance of MSLS-MA, comprehensive experiments were conducted on the MSTSP multi-solution optimization test suite. Experimental results demonstrate that our algorithm outperforms other comparative algorithms. Furthermore, to validate its practical utility, the algorithm was applied to solve instances from the TSPLIB benchmark library, successfully obtaining a set of significantly diverse and high-quality near-optimal solutions.

1. Introduction

The TSP stands as one of the most widely applied combinatorial optimization problems, underpinning numerous engineering and academic challenges. For instance, it is fundamental to the Vehicle Routing Problem [1, 2], Job Shop Scheduling Problem [3, 4], and the Block Relocation Problem [5]. In the classical TSP, a salesman departs from a city, visits each city exactly once, and returns to the origin. The objective is to identify the shortest possible tour. To address diverse practical requirements, researchers have proposed various TSP variants, including MMTSP [6, 7], multiobjective TSP [8, 9], and multimodal multiobjective TSP [10]. While most TSP applications and variants focus on locating a single optimal solution, the potential existence of multiple high-quality solutions is often overlooked.

However, providing distinct optimal alternatives to DMs is crucial in practice for several reasons. 1) Enhancing Solution Flexibility: Multiple solutions empower DMs to respond swiftly to unforeseen events (e.g., traffic congestion due to roadworks or flight cancellations caused by severe weather); 2) Improving Urban Road Network Capacity: By guiding DMs to choose different routes, multiple solutions distribute traffic flow, enhance the overall traffic capacity of urban roads, and achieve balanced traffic load distribution; 3) Accommodating Diverse Decision Preferences: Beyond explicit objective functions, DMs often have unstated constraints or subjective preferences; solutions should possess the capability to accommodate such implicit requirements. These requirements underscore the significance of research into the MMTSP.

The MMTSP is not merely a straightforward extension of the classical TSP. Rather, it constitutes an essential response to the complex transportation demands of the real world. It transcends the theoretical boundaries of single-modal optimization, providing an indispensable theoretical framework and solution methodology for addressing practical route planning problems involving coordinated multi-route selection, complex environmental constraints, and the pursuit of comprehensive benefits. Neglecting the critical dimension of multiple route alternatives and interchangeability will render optimization results practically irrelevant, failing to meet the growing demand for refined decision-making in modern logistics, transportation planning, and smart city development.

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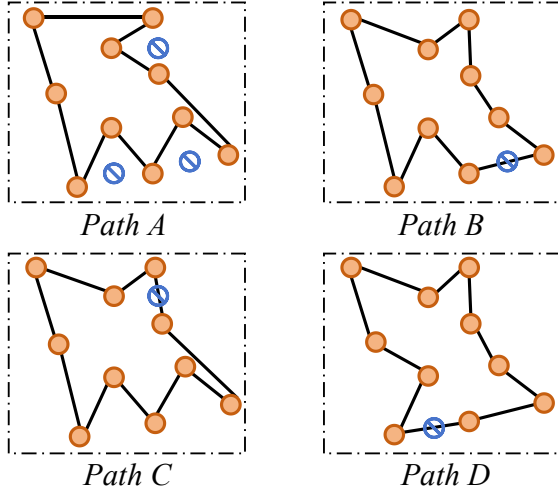


Figure 1: Illustration of the Multimodal Traveling Salesman Problem (MMTSP), where there are multiple equivalent globally optimal paths (paths A, B, C, and D), enabling a quick switch to an adjacent path when an obstacle appears.

As illustrated in Fig. 1, the MMTSP provides the DM with multiple feasible paths of equal length (e.g., Path A, B, C, D). When obstacles emerge in the environment, rendering some paths impassable (e.g., Paths B, C, D), the DM requires the capability to adjust the route to circumvent the obstruction dynamically. For single-modal TSP algorithms, this situation necessitates resolving the entire problem from scratch. In contrast, the multiple alternative solutions pre-computed by the MMTSP effectively empower the DM to perform rapid and flexible path switching. This scenario involving dynamic route replanning due to sudden obstacles is highly prevalent in real-world applications, underscoring the significant research significance of the MMTSP.

Solving the MMTSP relies on advances in multimodal intelligent optimization algorithms. These methods, designed to efficiently locate multiple global optima or local optima within an optimization problem, have garnered significant research interest in recent years [11]. To address this, numerous researchers have proposed various mechanisms to tackle such problems, including the Niching mechanism [12, 13], the Crowding Distance mechanism [14, 15], the Archive mechanism [16], the Clustering mechanism [17], the Indicator mechanism [18, 19], and the Machine learning mechanism [20, 21], among others. These mechanisms aim to balance the algorithm's exploration and exploitation capabilities to identify and maintain multiple optimal solutions within the decision space.

However, current multimodal optimization algorithms still face several critical challenges. Firstly, their convergence performance is often inadequate, hindering their applicability to practical problems. Unlike conventional intelligent optimization algorithms, multimodal approaches typically incorporate proactive exploration mechanisms to cover more decision space. This may lead to diminished exploitation capabilities, consequently making it difficult

to stably converge to the optimal solution within a limited number of function evaluations. Secondly, existing multimodal optimization problem (MMOP) test suites in the literature are predominantly confined to low-dimensional decision spaces. However, such low-dimensional problems exhibit insufficient complexity for general-purpose intelligent optimization algorithms, their true optimal solutions can often be efficiently obtained in very few generations, making it difficult to adequately evaluate an algorithm's multimodal handling capabilities [22]. Thirdly, the objective of requiring algorithms to simultaneously locate all global and local optima presents an inherent conflict with the primary goal of general intelligent optimization algorithms, which is to efficiently find a single high-quality global optimum. Fourthly, the majority of current research in the multimodal domain focuses on continuous domains, while investigations into discrete multimodal problems remain limited [23].

To address the aforementioned limitations, we propose a Multi-Strategy Local Search Memetic Algorithm for solving the MMTSP. MSLS-MA aims to obtain all global optimal solutions while balancing the exploration and exploitation capabilities of the multimodal algorithm. In summary, the main contributions are as follows:

- This paper proposes MSLS-MA, a multimodal memetic algorithm incorporating a diversity-driven initialization mechanism, an adaptive niching technique, and a population restart strategy. These components collectively enhance the algorithm's global exploration capability. Concurrently, the integration of a dedicated local search operator significantly strengthens its local exploitation capability. Consequently, MSLS-MA effectively locates multiple global optimal solutions.
- This paper proposes a two-stage evolutionary framework that divides the population evolution process into two stages: exploration and exploitation. This division effectively enhances the robustness of the algorithm in global search and local optimization. Therefore, the MSLS-MA algorithm can simultaneously solve TSP and stably obtain high-quality approximate global optimal solutions in practical application scenarios.
- The performance of MSLS-MA has been validated on various types of multimodal traveling salesman test problem sets [24]. According to the design method, MSTSP is divided into three categories: simple MSTSP, geometric MSTSP, and composite MSTSP. Experimental results show that MSLS-MA outperforms six state-of-the-art discrete multimodal intelligent optimization algorithms on most of the selected test problems. Additionally, to validate its practical application value, the algorithm was also applied to solve instances from the TSPLIB benchmark library, successfully obtaining a set of different and high-quality approximate optimal solutions.

The remainder of this study is structured as follows. Section 2 briefly reviews related works in the literature. Subsequently, Section 3 details the proposed MSLS-MA methodology. Section 4 then presents an experimental evaluation of MSLS-MA's performance, utilizing several multimodal evolutionary algorithms and established MMTSP test suites. Following this, Section 5 provides a comprehensive analysis, including an examination of parameter ε , the contributions of MSLS-MA's subcomponents, and the distinct search behaviors observed during its two-stage process. Finally, Section 6 concludes the paper and outlines potential avenues for future research.

2. Preliminary work

2.1. Multimodal optimization

Multimodal intelligent optimization methods aim to identify multiple global or local optimal solutions in optimization problems and have attracted widespread attention in recent years. Based on decision variable types, these problems are classified into continuous and discrete multimodal optimization. As shown in Fig. 2, continuous multimodal optimization problems have multiple distinct solutions (e.g., solutions x_1 , x_2 , and x_3) in the decision space that map to the same objective value. For continuous problems, first- or second-order derivatives are commonly used to identify critical points and guide search directions [25].

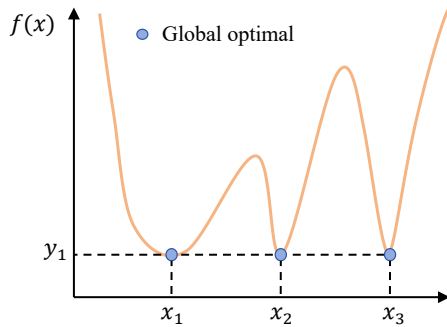


Figure 2: Illustration of a Continuous Optimization Problem, where multiple globally optimal solutions (x_1 , x_2 , x_3) exist in the decision space, sharing the same optimal objective function value.

Unlike continuous optimization, discrete optimization problems have decision variables from finite or countably infinite discrete sets, making their solution space discrete. This discreteness means no intermediate states exist between variable states, and transitions are jumps rather than continuous changes. As shown in Fig. 3, solutions x_2 and x_3 may be geometrically close but have significantly different variable values. For example, in TSP, solutions represent paths visiting all cities once, where objective values depend on overall path arrangement rather than incremental variable changes.

Discrete optimization problems have solution spaces composed of finite or countably infinite discrete decision variables. These problems are typically NP-hard, making it

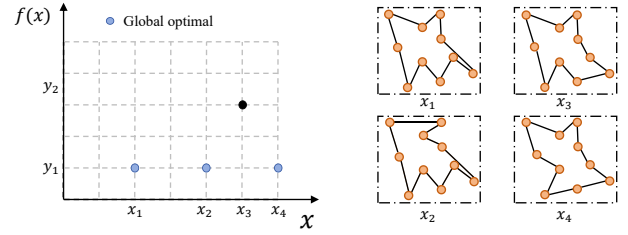


Figure 3: Illustration of the MMTSP, where the decision variables encode a Hamiltonian cycle, and multiple globally optimal solutions (x_1 , x_2 , x_4) exist in the solution space, sharing the same shortest path length.

computationally infeasible to obtain global optimal solutions within polynomial time [26]. Therefore, heuristic methods or exact algorithms are used to find high-quality feasible solutions. However, exact algorithms suffer from exponential computational complexity growth with problem dimension, limiting their applicability to large-scale instances [6]. In contrast, population-based evolutionary algorithms show significant advantages for discrete optimization by directly operating on natural discrete representations and efficiently achieving global exploration and local exploitation through selection, crossover, and mutation operators.

Evolutionary algorithms for solving MMOPs can be viewed as exploration and exploitation processes in the decision space [27, 28]. Generally speaking, overemphasizing exploration finds more feasible solutions but slows convergence, while overemphasizing exploitation accelerates convergence but risks local optima. To identify multiple optimal solutions in discrete optimization, researchers introduced niche techniques [29]. This approach divides the population into subpopulations, each converging to different potential optima while maintaining diversity through specific mechanisms. Since Goldberg [30] first proposed the preselection strategy, various niche techniques have emerged. Crowding and fitness sharing strategies are particularly widespread: crowding controls population updates by comparing similarity between offspring and parents, while fitness sharing reduces individual fitness based on proximity in solution space to regulate population density and maintain diversity.

In addition, multi-population strategies are also widely used, such as the Species Conservation Genetic Algorithm (SCGA) [31]. For clustering strategies, Qu et al. [32] proposed neighborhood mutation strategies (NCDE and NSDE) executing DE operators within Euclidean neighborhoods. Cheng et al. [33] developed a Q-learning-based brainstorming optimization algorithm (QLBSO2N) with two niche strategies and learning-assisted evolution to enhance global search and convergence. Huang et al. [34] proposed an adaptive differential evolution algorithm (WSNADE) using adaptive niche technology and protective two-scale mutation for improved diversity and search efficiency. Yan et al. [35] developed a knowledge-guided adaptive dynamic multimodal differential evolution utilizing self-organizing

mapping for population clustering and dynamic response mechanisms. Ji et al. [36] proposed a heterogeneous ensemble agent-assisted interval multimodal particle swarm optimization (IMPSON-HES) with agent model pools and incremental management to reduce function evaluations in high-cost optimization problems.

Although these methods have shown significant effectiveness in handling MMOPs, they still have some limitations. On the one hand, as the number of global optimal solutions for MMOPs increases, the possibility of these methods finding all global optimal solutions gradually decreases. On the other hand, for high-dimensional and highly complex problems, these fixed methods are almost unable to solve them effectively.

2.2. Memetic algorithms

Inspired by Darwin's evolutionary theory, Moscato [37] pioneered the Memetic Algorithm, which has been widely applied in human-machine collaborative scheduling [38], flowshop scheduling [39, 40], and job shop scheduling [41, 42]. MA is a hybrid optimization algorithm integrating global search (through crossover and mutation) with local search operators. Crossover and mutation emphasize exploration for broad decision space search, while local search focuses on exploitation within known solution neighborhoods.

For the exploration component, traditional genetic operators often generate ineffective gene exchanges, causing slow convergence. Liu et al. [43] designed elite gene extraction operators to guide neighborhood search and avoid redundant exploration. Guo et al. [44] proposed hybrid crossover operators for more efficient decision space exploration.

The exploitation component design depends on problem-specific characteristics. Zhai et al. [45] developed three drone collaboration operators for TSP with drones. Li et al. [46] created four problem-oriented local search operators for energy-efficient distributed flexible job shop scheduling. Lu et al. [38] constructed five variable neighborhood search operators for human-machine collaborative scheduling.

Balancing exploration and exploitation is a key MA challenge. Liu et al. [47] introduced dynamic step-size adjustment using historical information. Li et al. [48] employed Tchebycheff decomposition-based selection to balance convergence and diversity. Zou et al. [49] developed dual-individual population mechanisms for collaborative evolution. Wang et al. [50] proposed a Penalized Boundary Intersection decomposition-based solution selection strategy that effectively balances convergence and diversity. Traditional Memetic Algorithm, however, suffer from suboptimal search dynamics due to their fixed local search mechanisms, which typically overemphasize exploitation during early generations while providing insufficient exploration in later stages. Zhang et al. [51] divided evolution into three stages: global search, hybrid search, and local search to better balance exploration and exploitation in their Time-adaptive Memetic Algorithm.

2.3. Traveling salesman problem

The TSP is a combinatorial optimization problem defined on graph $G = (V, E)$, where V and E are vertex and edge sets respectively. The objective is finding a Hamiltonian circuit visiting each vertex exactly once while minimizing total path length. Traditional TSP research focuses on finding single optimal solutions, while MMTSP aims to simultaneously identify multiple optimal paths for decision-makers to adapt to different circumstances. MMTSP emphasizes solution diversity and balance. Both single-modal and MMTSP share the same mathematical expression:

$$\min f(\pi) = \sum_{n=1}^{N-1} c_{\pi(n)\pi(n+1)} + c_{\pi(N)\pi(1)} \quad (1)$$

Here, c_{ij} represents the path from city i to city j , and $\pi(n)$ represents the n th city in path π .

TSP research using evolutionary algorithms began in the 1980s with representative algorithms like genetic algorithms (GA) [52] and ant colony algorithms (ACA) [53]. Early studies proposed classical genetic operators including ordered crossover [54], partial mapping crossover [55], and cyclic crossover [56]. However, these operators often underperformed compared to other heuristic methods in computational efficiency. Subsequently, researchers developed more efficient operators such as edge recombination crossover [57], edge assembly crossover [58, 59], and improved cyclic crossover [60].

Maintaining decision space diversity has proven key to improving TSP solution efficiency using evolutionary algorithms [61, 62]. Akhand et al. proposed the Discrete Spider Monkey Optimization algorithm [63], introducing Swap Sequence and Swap Operator with grouping update mechanisms for local and global leaders, significantly improving TSP performance. Zheng et al. [64] designed TL-PSO with adaptive crossover and clustering-guided mutation to enhance search capability. Zhang et al. [65] developed GA-JGHO, effectively maintaining population diversity through improved roulette wheel selection and jump gene operators.

Evolutionary computation focuses on solving MMOPs [66, 67]. Several multimodal optimization algorithms have been developed for MMTSP, including Niche-based Ant Colony System (NACS) [68], Neighborhood-based Genetic Algorithm (NGA) [24], Niching Memetic Algorithm (NMA) [6], and Niching Regression Adaptive Memetic Algorithm (NRAMA) [69]. These algorithms employ distinct strategies to maintain population diversity and locate multiple optimal solutions.

NACS utilizes multiple pheromone matrices for different solution regions, creating new matrices when ants discover superior solutions. However, it faces challenges in niche parameter configuration without prior problem knowledge. NGA clusters neighboring individuals into subpopulations for standard genetic operations to locate diverse local optima, but suffers from slow convergence and difficulty defining appropriate neighborhood sizes. NMA clusters similar solutions into adaptive neighborhood groups, incorporating critical edge awareness and selective local search,

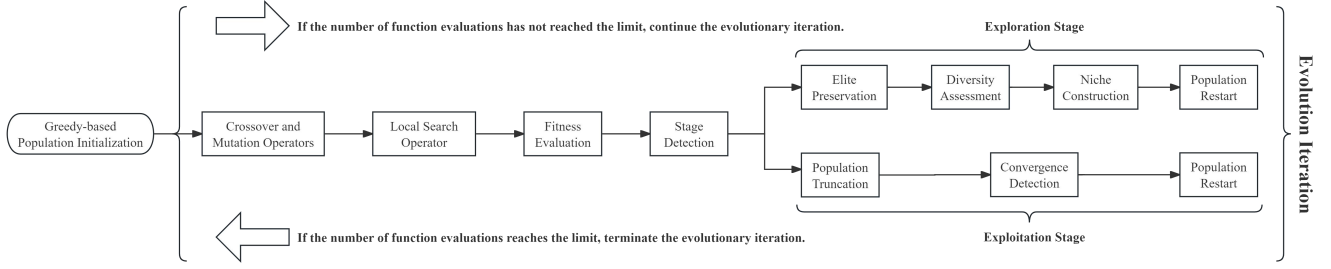


Figure 4: Diagram of the Two-Stage Evolutionary Framework of MSLS-MA, utilizing crossover, mutation, and local search operators to explore the decision space, with stage selection determined by function evaluation expenditure.

yet consumes high computational resources and performs poorly on large-scale instances. NRAMA employs regression partitioning initialization and neighborhood clustering with adaptive parameter control and redundant individual concepts, enhancing efficiency for large-scale TSP instances.

Discrete multimodal optimization algorithm development focuses on three aspects: 1) efficient diversity preservation mechanisms; 2) rapid convergence capabilities; 3) robust parameter control strategies.

3. Multi-Strategy Local Search Memetic Algorithm for MMTSP

3.1. Framework

Acquiring well-distributed multiple optimal solutions is one of the core objectives of multimodal evolutionary algorithms. To achieve this, existing multimodal evolutionary algorithms typically embed crowding distance in the decision space within their environmental selection strategies. For low-dimensional problems, such methods are highly effective in obtaining multiple equivalent optimal solutions. However, a conflict exists between convergence in the objective space and diversity in the decision space. Overemphasizing decision space diversity can compromise convergence for high-dimensional problems.

Inspired by literature [70, 71], this paper proposes a novel two-stage evolutionary mechanism to better balance convergence and diversity in the decision space. As illustrated in Fig. 4, the mechanism first applies crossover and mutation operators to implement global search across the population, followed by local search operators for neighborhood exploration. The framework adaptively selects the evolutionary stage based on the number of function evaluations already consumed: during the early stage of evolution, an exploration strategy is employed during the early evolutionary stage, utilizing a larger niche radius to explore a broader decision space; during the later evolutionary stage, the mechanism shifts to a convergence strategy, ceasing niche-based exploration of the decision space and instead utilizing local search for finer-grained exploitation of the population. This mechanism effectively enhances convergence during the evolutionary process, enabling the acquisition of as many optimal solutions as possible. It also

Algorithm 1 General Framework of MSLS-MA

Population size N

Input: Maximum function evaluations $MaxFEs$

Maximum stagnation count $MaxStagnation$

Output: Optimal path and solution $pspf$

```

1:  $Pop \leftarrow \text{Initialization}(N)$ 
2:  $Fitness \leftarrow \text{CalFitness}(Pop)$ 
3: while  $count \leq MaxFEs$  do
4:    $Parent \leftarrow \text{Crossover}(Pop)$ 
5:    $Off \leftarrow \text{Mutation}(Parent)$ 
6:    $Pop \leftarrow [Off; Pop]$ 
7:    $[Pop, count] \leftarrow \text{Localsearch}(Pop, Off, count)$ 
8:    $Fitness \leftarrow \text{CalFitness}(Pop)$ 
9:    $Stage \leftarrow \text{StageJudgment}(count, MaxFEs)$ 
10:   $Pop \leftarrow \text{EnvSelection}(Pop, Stage)$ 
11:   $count \leftarrow count + \text{size}(Pop)$ 
12:   $Pop \leftarrow \text{PopulationRestart}(Pop, Stage)$ 
13: end while
14:  $pspf \leftarrow \text{Post-processing}(Pop)$ 
  
```

holds promise for extension to other multimodal evolutionary algorithms. The pseudocode for the entire MSLS-MA framework is described in Algorithm 1.

3.2. Environmental selection

For the MMTSP, identical objective function values may correspond to distinct solution paths. Therefore, if a convergence-prioritized strategy is naively employed to select solutions from the combined population, the algorithm will struggle to adequately explore the decision space and is likely to become trapped in local optima. Clearly, we aim to retain higher-performing solutions while simultaneously maintaining population diversity in the decision space, but these objectives are often conflicting.

To address this issue, we first adopt a niching selection mechanism to explore the decision space. By constructing multiple niches, we retain individuals exhibiting better diversity within each niche and remove similar individuals. This ensures that the next generation preserves the best solutions found thus far while maintaining population diversity. Subsequently, after removing similar solutions, individuals are selected based on both tour length and diversity level.

Algorithm 2 Environmental Selection

Population size N
Joint population Pop
Input: Niching radius $NichingRadius$
Diversity threshold $DiversityThreshold$
Output: Updated population Pop

```

1:  $Pop \leftarrow NichingSelection(Pop, NichingRadius, DiversityThreshold, N)$ 
2:  $Pop \leftarrow SortRows(Pop, N + 2)$ 
3: if  $Num(Pop) > N$  and  $count \leq \epsilon \times MaxFEs$  then
4:    $elite \leftarrow Pop(1 : \min(NumElite, size(Pop, 1)))$ 
5:    $RemainingSolution \leftarrow Pop(NumElite + 1 : \text{end})$ 
6:    $Remain \leftarrow DiversityScores(RemainingSolution)$ 
7:    $Selected \leftarrow RemainSortIdx(1 : \min(N - NumElite, length(sortIdx)))$ 
8:    $Pop \leftarrow [elite; Selected]$ 
9: end if
10: if  $Num(Pop) > N$  and  $count > \epsilon \times MaxFEs$  then
11:    $Pop \leftarrow sortrows(partial, N + 2)$ 
12:    $Pop \leftarrow partial(1 : N, :)$ 
13: end if

```

To enhance niche construction and refinement, an adaptive approach is integrated into the algorithm. This approach dynamically adjusts the parameters and methodology of niching selection according to the evolutionary progress. The environmental selection process is depicted in Algorithm 2.

Specifically, the parent population and its offspring population are first combined to form a joint population. Subsequently, niche computation and local search operations are performed on the joint population (Lines 1-2). Depending on the current evolutionary stage of the population (exploration or exploitation), an appropriate individual selection strategy is applied. During the exploration stage, solutions are prioritized for retention based on their diversity metric (i.e., the degree of difference between individuals), favoring those exhibiting higher diversity (Lines 4-10). In contrast, during the convergence stage, a truncation selection method based on objective value is employed to maintain the fixed population size (Lines 11-14).

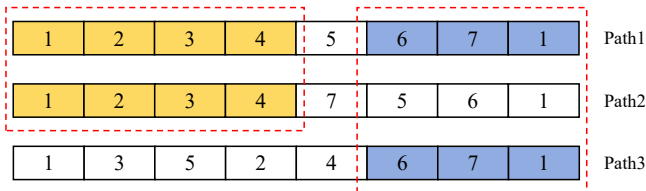


Figure 5: Illustration of Path Similarity, where the marked paths represent identical paths, and the length of the marked paths corresponds to the similarity score.

To effectively quantify the differences between individuals within a population, this paper constructs a tour

similarity metric. This metric evaluates the degree of similarity between solutions by counting the number of differing segments in their path sequences. As shown in Fig. 5, path 1 represents the current population's optimal solution, while path 2 and path 3 are examples of other individuals within the population.

3.3. Local search operator

Local search is an optimization method based on neighborhood structures, which improves solution quality by iteratively searching for better solutions within the neighborhood of the current solution. In TSP research, the 2-opt [72] and 3-opt [73] operators are the most widely applied local search operators. The 2-opt operation removes two non-adjacent edges, splitting the current tour into two separate segments, and then reconnects these segments differently to construct a new feasible tour. Similarly, the 3-opt operation involves removing and replacing three edges to generate a new tour.

However, traditional approaches typically perform local search on every solution in each generation's population, resulting in the algorithm consuming a substantial number of objective function evaluations [74]. To enhance efficiency, Li et al. [75] proposed a neighborhood-based local search strategy. This strategy performs operations only on strongly correlated nodes within the neighborhood, effectively reducing computational overhead.

The core mechanism of local search algorithms lies in defining the neighborhood generation rules and neighborhood selection strategies. As illustrated in Fig. 6, the 2-opt operator generates a new tour by reversing the order of the subpath between any two nodes in the current tour. Conversely, the 3-opt operator selects three breakpoints to divide the tour into four segments (A, B, C, D), then constructs a new tour by reversing selected segments and modifying the connection sequence between them. To enhance algorithmic performance, this paper integrates both the 2-opt and 3-opt operators. The specific implementation is detailed in Algorithm 3.

The procedure operates as follows. First, an iteration counter is initialized. While improvement potential exists and the maximum iteration count is not exceeded, a nested loop traverses all possible combinations of apath indices. For each segment $[i+1, j]$, a reversal operation generates a new candidate tour. When the intensity value exceeds the limit and the iteration limit has not been reached, an inner loop is activated. This inner loop iteratively applies the 3-opt optimization operator, employing an edge exchange strategy to refine the tour. The path state and counters are updated after each operation. This process continues until the termination criteria are satisfied.

3.4. Population restart

When applying multimodal evolutionary algorithms to solve the MMTSP, populations are prone to premature convergence. To enhance the algorithm's ability to escape local optima, we design a Dynamic Convergence Detection Mechanism. After multiple iterations without improvement,

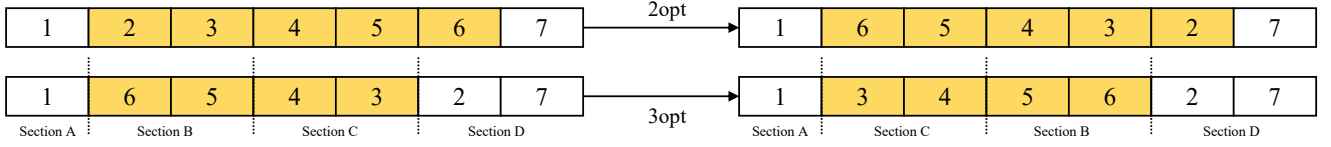


Figure 6: Illustration of classic local search techniques: The 2-opt operator generates new solutions by reversing the subpath between any two nodes. The 3-opt operator, in contrast, selects three breakpoints, disconnecting the tour into four segments, and constructs a new tour by reversing selected segments and recombining the remaining parts.

Algorithm 3 Local Search Procedure

Population size N
 Elite population Pop
Input: Path length matrix c_{ij}
 Search intensity $intensity$
 Evaluation count $count$
Output: Updated elite population Pop
 Updated evaluation count $count$

```

1:  $iteration \leftarrow 0$ 
2:  $improved \leftarrow true$ 
3:  $maxIteration \leftarrow N * intensity$ 
4: while  $improved$  and  $iteration < maxIterations$  do
5:    $improved \leftarrow false$ 
6:   for  $route \in Pop$  do
7:      $[route, count, improved] \leftarrow 2opt(route, N, c_{ij}, count, MaxFEs)$ 
8:    $iteration \leftarrow iteration + 1$ 
9:   end for
10: end while
11: if  $intensity > 0.5$  and  $iteration < maxIteration$  then
12:    $improved \leftarrow true$ 
13:   while  $improved$  and  $iteration < maxIteration$  do
14:      $improved \leftarrow false$ 
15:     for  $route \in Pop$  do
16:        $[route, count, improved] \leftarrow 3opt(route, N, c_{ij}, count, MaxFEs)$ 
17:      $iteration \leftarrow iteration + 1$ 
18:     end for
19:   end while
20: end if
```

we determine that the population has entered a local convergence state.

During the exploration stage of population evolution, we implement an elite preservation strategy. Elite individuals from the current population are archived in an elite repository. Simultaneously, a diversity-enhanced population initialization algorithm generates new individuals to replace low-fitness solutions, thereby maintaining population diversity. This strategy ensures the inheritance of the current optimal solutions while effectively expanding the search space through the introduction of exploratory solutions. It achieves a balance between the algorithm's exploitation and exploration capabilities.

Algorithm 4 Population Restart Mechanism

Population Pop
 Population size N
 Path length matrix c_{ij}
Input: Current stagnation count SC
 Maximum stagnation count MSC
 Path length matrix $cost$
Output: Updated population Pop

```

1:  $s \leftarrow Pop(:, end)$ 
2:  $Pop \leftarrow Pop(:, 1 : end - 1)$ 
3: if  $\min(s) \geq BestFitness$  then
4:    $SC \leftarrow SC + 1$ 
5: end if
6: if  $SC \geq MSC$  and  $count \leq \epsilon \times MaxFEs$  then
7:    $ElitePop \leftarrow Pop(SortIdx(1 : NumElite), :)$ 
8:    $newPop \leftarrow ImprovedInitialization(N, N - NumElite, c_{ij})$ 
9:    $Pop \leftarrow [ElitePop; newPop]$ 
10: end if
11: if  $SC \geq MSC$  and  $count > \epsilon \times MaxFEs$  then
12:   for  $i = 1$  to  $NumElite$  do
13:      $[Pop(i, :), count] \leftarrow LocalSearch(Pop(i, :), c_{ij}, intensity, count, MaxFEs, N)$ 
14:   end for
15: end if
```

During the exploitation stage of population evolution, a local search strategy is applied to elite individuals within the current population. This strategy conducts intensive local searches on elite solutions, assisting the population in escaping local optima and consequently enhancing overall optimization performance.

To mitigate population stagnation during evolution, a stagnation detection mechanism is proposed in this work. The mechanism monitors the quality and quantity of optimal solutions over consecutive generations to identify stagnation states. Upon detecting stagnation within a predefined generation window, the algorithm employs stage-specific restart strategies based on the current evolutionary phase.

Fig. 7 illustrates the differing effects of the population restart strategy across evolutionary stages. During the exploration stage, to fully explore the solution space and obtain diversity information, a strategy is adopted to retain most of the elite solutions, and the population is initialized and restarted based on a greedy algorithm. This mechanism randomly

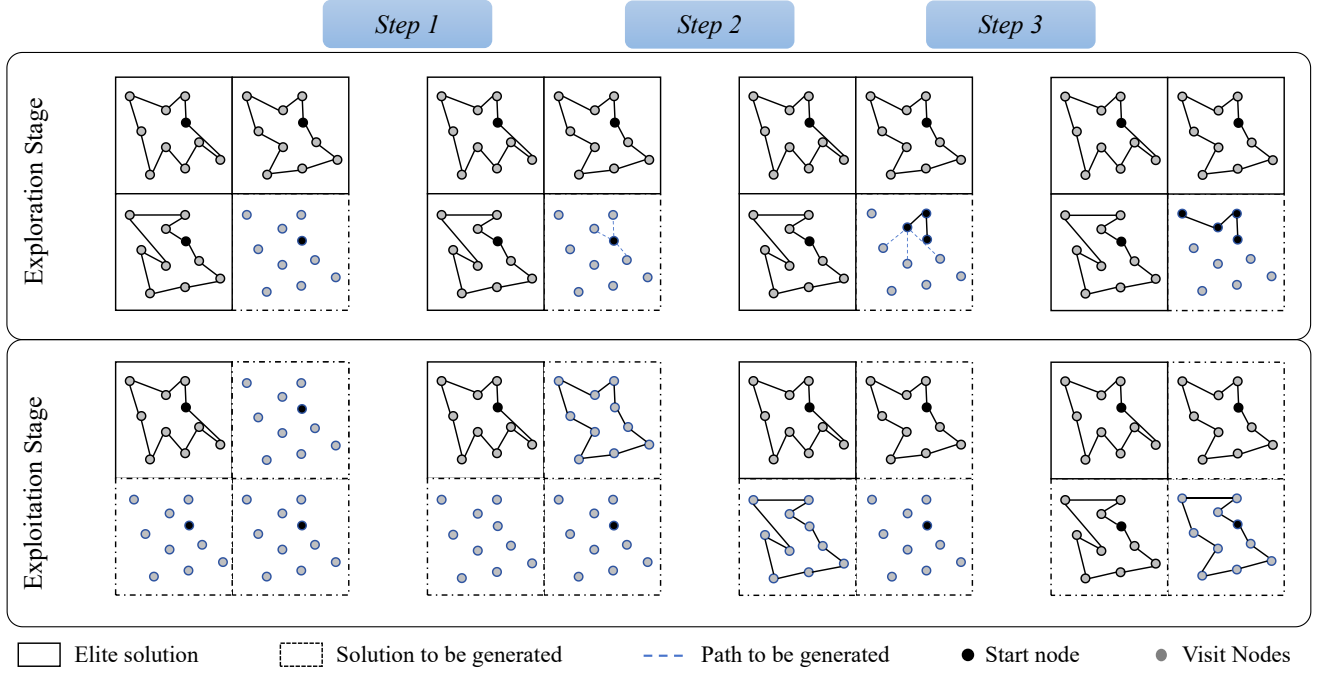


Figure 7: Illustration of Population Restart. During the exploration stage, the restart operation retains a significant number of elite individuals while introducing new ones, aiming to maintain population diversity; whereas in the exploitation stage, it retains only a small number of elite individuals and subjects them to intensive local search, with the goal of enhancing the algorithm's convergence precision.

generates feasible solutions to promote a broad search of the solution space. In contrast, during the exploitation stage, the population's objective shifts to accelerating convergence. At this point, the strategy switches to a local search restart mechanism that retains a small number of elite solutions, using existing information to guide the population toward the optimal solution region.

4. Experiment

4.1. Experimental setting

4.1.1. Competitor algorithms

To evaluate the performance of the MSLS-MA algorithm on the MMTSP, this study selected six effective evolutionary algorithms as benchmark comparison algorithms, including: genetic algorithm (GA) [52], crowding genetic algorithm (CGA) [30], Sharing Genetic Algorithm (ShGA) [30], Niche Adaptive Genetic Algorithm (NAGA) [76], Niche Genetic Algorithm (NGA) [24], and Niche Memory Algorithm (NMA) [6]. Specifically, GA serves as the base algorithm. CGA and ShGA are variants of GA, respectively incorporating a crowding distance mechanism and an adaptation degree sharing mechanism to maintain population diversity. NAGA further integrates a niche-scale adaptive mechanism and a dynamic peak identification mechanism based on ShGA. Meanwhile, NMA and NGA are algorithms specifically designed to solve multimodal traveling salesman problems.

To ensure fairness in the algorithm comparison, the population size N for all algorithms was set to 150. The maximum number of function evaluations was dynamically adjusted based on the type of test problem, set to 60,000 for the first 12 test problems and 1,200,000 for the remaining 13 test problems. This parameter setting scheme is based on the literature. Additionally, all evolutionary algorithms use simulated binary crossover and polynomial mutation algorithms to generate offspring. The specific parameters for each comparison algorithm are set according to the original literature. For MSLS-MA, the hyperparameter for algorithm stage transition is set to 0.4, meaning that when the number of evaluations is less than or equal to $0.4 \times \text{MaxFEs}$, the algorithm is in the exploration stage, and when the number of evaluations is greater than $0.4 \times \text{MaxFEs}$, the algorithm is in the exploitation stage. All computational experiments were conducted on a desktop computer equipped with an AMD Ryzen 9 5900X 12-core processor and 64 GB of RAM, utilizing MATLAB R2023b as the programming environment. To ensure statistical reliability and account for the stochastic nature of the algorithms, each algorithm was executed independently for 30 runs.

4.1.2. Benchmark

To comprehensively evaluate the performance of the MSLS-MA algorithm, this study adopted two types of benchmark test sets: the MSTSP benchmark test set [24] and the TSPLIB standard test set [77]. The MSTSP benchmark

test set contains 25 problem instances, which are further categorized into three types: simple MSTSP, geometric structure MSTSP, and composite MSTSP. At the same time, in order to examine the performance of the algorithm in solving practical problems, we selected five TSP problem instances from the TSPLIB standard test set for evaluation.

4.1.3. Performance metrics

For MMOPs, two primary evaluation metrics are employed: F_β and Diversity Indicator (DI) [24]. For completeness, we present the following detailed specifications:

F_β measure: F_β is a composite metric integrating precision (P) and recall (R), employed to evaluate the quality of solutions obtained.

$$P = \frac{TP}{TP + FP} \quad (2)$$

where TP denotes the number of correctly identified optimal solutions among the solutions obtained by the algorithm, and FP represents the number of erroneously identified non-optimal solutions within the solution set.

$$R = \frac{TP}{TP + FN} \quad (3)$$

where FN denotes the number of undetected optimal solutions. Note that the cumulative count of TP and FN equals the total number of ground-truth optimal solutions in the benchmark set. The F_β metric [78] is then formulated as:

$$F_\beta = \frac{(1 + \beta^2) \times P \times R}{\beta^2 \times P + R} \quad (4)$$

In evaluation metrics, the β^2 parameter is conventionally set to 1, assigning equal weight to precision (P) and recall (R). However, in scenarios where test instances contain numerous optimal solutions, identifying representative solutions becomes more critical than exhaustively enumerating all solutions. Based on this consideration, this study sets $\beta^2 = 0.3$ to enhance the weight of precision in the composite evaluation, thereby better capturing the representative quality of obtained solutions. Furthermore, the three adopted evaluation metrics precision (P), recall (R), and the composite metric (F_β) all yield real-valued results within the $[0, 1]$ range. Specifically, two boundary cases exist: 1) F_β achieves its maximum value of 1 when the algorithm identifies all desired solutions under ideal conditions; 2) F_β attains its minimum value of 0 when the algorithm fails to identify any valid solutions.

DI: The DI serves as another critical performance measure for algorithm evaluation. When all algorithms fail to locate desired solutions (yielding identical $F_\beta = 0$ values), DI provides further discrimination capability among algorithmic performance. Drawing inspiration from evolutionary multi-objective optimization principles, DI quantifies the diversity performance of solution sets ultimately provided

by algorithms. This quantitatively reflects how extensively the solution set converges toward distinct optimal solutions within the ground-truth optimal solution set (P). Formally, DI is defined based on the average maximum similarity between solutions in the obtained set and individuals in the reference set (P), with its computational formulation given by:

$$DI(\mathbb{P}, \mathbb{S}) = \frac{\sum_{i=1}^{|\mathbb{P}|} \max_{j=1, \dots, |\mathbb{S}|} S(p_i, s_j)}{N \times \|\mathbb{P}\|} \quad (5)$$

where $\mathbb{P}$ denotes the ground-truth solution set and $\mathbb{S}$ represents the algorithmically obtained solution set. The expression $\sum_{i=1}^{|\mathbb{P}|} \max_{j=1}^{|\mathbb{S}|} S(p_i, s_j)$ computes the maximum similarity between each permutation $p_i \in \mathbb{P}$ and all permutations in $\mathbb{S}$, formally defined as follows:

$$S(\pi_i, \pi_j) = \frac{|\Phi(\pi_i) \cap \Phi(\pi_j)|}{N} \quad (6)$$

where $\Phi(\pi_i)$ and $\Phi(\pi_j)$ denote the edge set of π_i and π_j , respectively; $\Phi(\pi_i) \cap \Phi(\pi_j)$ means the intersection set of $\Phi(\pi_i)$ and $\Phi(\pi_j)$; and $|\cdot|$ denotes the number of edges in $\Phi(\pi_i) \cap \Phi(\pi_j)$. The sharing distance measure operates on the edge set of a route rather than the absolute node positions. So that solutions having different absolute node orders but the same visiting edge set are identified as the same solution.

4.2. Results

4.2.1. Performance on MSTSP test problems

To evaluate the algorithm's solution performance on the MMTSP, we conducted experiments using the standardized MSTSP benchmark set. This test suite comprises 25 test instances, with experimental results summarized as follows:

F_β : The F_β metric quantifies solution quality, with higher values indicating superior solutions. We conducted experimental comparisons of the proposed MSLS-MA algorithm against six baseline algorithms using F_β across all 25 MSTSP instances. As indicated in Table 1, the results demonstrate that MSLS-MA achieved significantly better F_β outcomes on all 25 test problems. In contrast, the six baseline algorithms exhibited inferior performance on complex MSTSP instances, particularly encountering difficulties when handling large-scale urban configurations. This limitation likely stems from their excessive evaluation budget consumption during decision space exploration, ultimately compromising convergence to optimal solutions. Conversely, MSLS-MA's two-stage framework effectively maintains the exploration-exploitation balance (diversification versus intensification).

DI: The DI metric assesses solution diversity, with higher DI values indicating greater diversity. As presented in Table 1, MSLS-MA and CGA demonstrate comparatively superior diversity performance across the complete MSTSP instance set. For simple-type and geometric-type MSTSP instances, GA, NMA, CGA, NGA, and NAGA exhibit satisfactory diversity performance. However, these

Table 1

The average rankings for each algorithm category on MSTSP test suites

Test Suites	Indicators	NMGA	ShGA	CGA	GA	NGA	NMA	MSLS-MA
Simple_MSTSP	F_β	6.83	6.00	1.33	1.00	5.17	1.50	1.00
	DI	7.00	6.00	1.33	1.00	5.00	1.50	1.30
	ET	5.00	3.00	1.67	4.00	-	-	1.33
Geometry_MSTSP	F_β	6.00	6.50	2.33	1.33	5.33	3.17	1.17
	DI	6.50	6.50	2.50	1.50	5.00	2.50	1.17
	ET	5.00	3.17	1.67	3.83	-	-	1.33
Composite_MSTSP	F_β	5.38	5.46	2.00	4.00	5.15	3.15	1.00
	DI	5.92	6.38	2.92	4.62	4.85	1.92	1.08
	ET	5.00	3.00	3.92	2.00	-	-	1.00

algorithms experience significant degradation in diversity on composite-type MSTSP instances. In contrast, ShGA and NAGA yield suboptimal diversity outcomes across all problem instances. Collectively, the evaluated algorithms achieve enhanced diversity results on simple-type MSTSP instances characterized by smaller decision variable spaces. Notably, MSLS-MA consistently demonstrates superior comprehensive performance across all evaluation metrics.

Execution Time: The execution time (ET) serves as another crucial metric for evaluating algorithm performance. For NP-hard problems, computational expense increases dramatically with city scale. As presented in Table 1 (optimal results highlighted in bold), MSLS-MA consistently demonstrated superior execution efficiency across the entire benchmark set. Specifically, MSLS-MA exhibited robust performance across all problem types, without exhibiting the significant runtime increases observed for CGA on simpleMSTSP and CompositeMMTSP instances. Taking MSTSP1 (9 cities) as an example, the average execution times (in seconds) for the algorithms, listed in ascending order, are: CGA (0.61), MSLS-MA (0.67), ShGA (0.81), GA (1.07), NMGA (1.19). For the larger-scale MSTSP25 (66 cities), the execution time ranking shifted: MSLS-MA (28.33), GA (44.32), ShGA (82.11), CGA (134.20), NMGA (218.43). This observation indicates that the execution times of NMGA, CGA, and ShGA escalate rapidly with increasing city size. Comparatively, the execution time of MSLS-MA is less affected by increases in city size.

4.2.2. Performance on TSPLIB test problems

Applying multimodal optimization algorithms to the TSP presents a fundamental challenge: the inherent multimodal properties of a target TSP instance cannot be predetermined. When addressing inherently unimodal problems, multimodal evolutionary algorithms may exhibit reduced efficiency or slower convergence rates. This challenge can be mitigated through two strategic approaches:

- **Optimality Guarantee:** Well-designed multimodal memetic algorithms, leveraging their potent capabilities for global exploration and local exploitation, possess the potential for asymptotic convergence to the global

Table 2

The shortest path of all compared algorithm on different test suites.

	Optimal	GA	CGA	MSLS-MA
eil51	426	435	444	426
1berlin52	7542	8167	7972	7542
st70	675	730	685	675
pr76	108159	110740	115244	108159
kroA100	21282	22294	22625	21282
lin105	14379	14754	15678	14379

optimal solution, regardless of the problem's modality determinacy.

- **Feasibility Guarantee:** Even when precise convergence to the global optimum is unattainable, the multimodal mechanism facilitates the discovery of multiple high-quality local optima. This provides valuable suboptimal alternatives for practical applications, thereby enhancing the algorithm's robustness and applicability.

To evaluate algorithmic performance on single-modal TSP, experiments were conducted using the standard TSPLIB benchmark repository comprising 112 TSP instances. Three algorithmsGA, CGA, and MSLS-MAwere applied to eight selected TSP instances. The maximum function evaluations were uniformly set to $50000 \times N$, where N denotes city count.

First, we assessed algorithms' capability to locate global optima. Table 2 summarizes the shortest tour lengths obtained by each algorithm, with optimal results boldfaced. As demonstrated, MSLS-MA achieved optimal tour lengths across all test instances, significantly outperforming both GA and CGA.

Then, to elucidate the multimodal characteristics of the solution space more explicitly, this study selected four exemplary solutions obtained by the MSLS-MA on the eil51 instance, visualized in Fig. 8. Each subfigure in Fig. 8 presents distinct locally optimal solutions discovered by MSLS-MA. These solutions share identical objective function values

yet exhibit divergent topological structures. These observations specifically, the existence of multiple solutions with identical objective values but different structures provide compelling evidence for the inherent multimodal properties of this TSP instance's solution space.

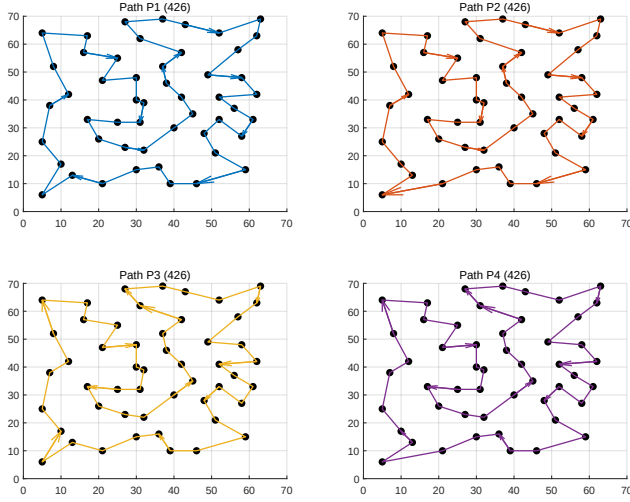


Figure 8: Schematic of exemplary solutions obtained by MSLS-MA for *eil51*, where the ground-truth optimal tour length is 426 versus the algorithmically obtained best length of 426.

The algorithm successfully obtained multimodal optimal solutions for the *eil51* test instance, where solution paths share identical lengths but exhibit distinct topological structures. As illustrated in Fig. 8, the *eil51* test problem demonstrates multimodal characteristics. Specifically, Paths 1, 2, 3, and 4 share identical node sequences across most segments, differing only in localized regions to manifest unique topological configurations.

4.2.3. Overall performance

As evidenced by Table 1 and Table 2, MSLS-MA demonstrates exceptional performance in multimodal domains. It achieves superior results in the F_β metric, attaining the highest values across all grouped test problems and securing first-place rankings in all 31 test instances. Concurrently, MSLS-MA exhibits comprehensive dominance in the DI metric, achieving optimal rankings throughout the test suite. In unimodal testing, MSLS-MA further demonstrates enhanced stability, reliably converging to acceptable solutions with greater consistency and outperforming comparative algorithms in overall performance. Collectively, these experimental findings establish MSLS-MA as a highly competitive multimodal evolutionary algorithm applicable to both multimodal and unimodal Traveling Salesman Problems.

5. Further Discussion

5.1. Analysis on parameter ϵ

In the supplementary material, we present experimental investigations into the parameter settings for the MSLS-MA

Table 3

The average rankings for each algorithm category on the MSTSP test suites, where V1V6 utilize distinct stage control parameters ϵ .

Test Suites	Ind.	V1	V2	V3	V4	V5	V6
Simple	F_β	1.00	1.00	1.00	1.00	1.00	1.00
	DI	1.00	1.00	1.00	1.00	1.00	1.00
Geometry	F_β	1.50	1.00	1.50	2.00	2.33	2.67
	DI	1.50	1.67	1.33	2.00	2.33	2.67
Composite	F_β	2.33	3.75	2.00	2.67	2.75	3.42
	DI	2.77	2.92	1.92	2.92	3.92	4.92

algorithm. Due to space limitations, only a concise summary is provided in the main text. As shown in supplementary material, the experiments revealed that: the population size should be set between 150 and 250, the stage adjustment parameter is recommended to lie between 0.2 and 0.4, and the elite local search proportion should be set within the range of 0.3 to 0.6. This section presents the configuration of stage adjustment parameter adjustments and their impact on algorithm performance. Detailed results data can be found in the supplementary material.

Table 3 illustrates the impact of the stage control parameter ϵ on algorithm performance. Analysis of the data within the table reveals that the optimal value range for ϵ is [0.2, 0.4]. On the simple test problem, the setting of ϵ exhibits no significant influence on the solution outcomes. This is primarily because the problem scale is small (low dimensionality), and even if the algorithm focuses solely on the exploration or exploitation stage, its local search operations can effectively locate all global optimal solutions. In contrast, on the geometry and composite test problems, the value of ϵ significantly affects the solution quality: an excessively low ϵ compromises exploration capability, hindering the discovery of potential optimal regions, while an excessively high ϵ weakens exploitation capability, preventing the algorithm from converging sufficiently to the global optimum. Notably, on the MSTSP test problem set, the algorithm still obtains relatively good solutions despite suboptimal parameter settings. This can be attributed to two main factors: first, the current MSTSP problem set lacks high-dimensional, complex instances, reducing parameter sensitivity; second, there are minimal differences between the optimal solutions within this problem set. Once the algorithm locates any one optimal solution, its local search operator can efficiently discover all other optimal solutions within its neighborhood.

5.2. Analysis on subcomponents of MSLS-MA

To investigate the impact of key sub-components on algorithm performance, this study developed three variants (MSLS-MA-V1 to V3) based on the baseline algorithm,

sequentially removing the dynamic niche strategy, the selective local search operator, and the population restart mechanism. Due to space limitations, detailed analyses are provided in Supplementary Material, only the main conclusions are summarized here.

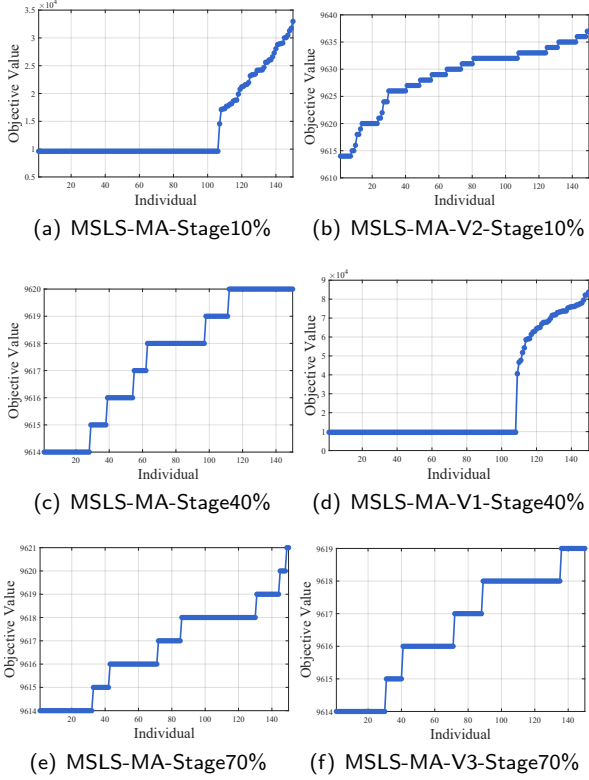


Figure 9: The schematic illustrates the effects of different components: subfigures (a) and (b) correspond to the effect of the niche component; subfigures (c) and (d) illustrate the impact of the local search operator; and subfigures (e) and (f) reflect the effect of the population restart strategy.

Fig. 9 illustrates the impact on algorithm performance resulting from the removal of specific components. Analyzing the individual distribution in the early evolutionary stage reveals the effect of the niching component. A comparison between subfigures (a) and (b) demonstrates that the proposed dynamic niching strategy effectively preserves population diversity by restricting the number of solutions within an individual's neighborhood during this stage. This mechanism underpins the algorithm's robust performance across various MSTSP instances. Examining the individual distribution in the mid evolutionary stage highlights the influence of the local search operator. Comparing subfigures (c) and (d) shows that the selective local search operator conducts a thorough exploration of an individual's neighborhood, leading to a significant enhancement in the algorithm's convergence efficiency. Observing the individual distribution in the late evolutionary stage demonstrates the role of the population restart component. Contrasting subfigures (e) and (f) indicates that the population restart strategy enables the algorithm to effectively escape local optima without

significantly compromising the quality of the final solution set.

5.3. Analysis on MSLS-MA's searching behaviors

This study proposes a two-stage evolutionary framework that employs a differentiated evolutionary strategy, aiming to optimize the balance between exploration and exploitation during population search. To provide an in-depth analysis of the search behavior characteristics of the algorithm across different evolutionary stages, the benchmark instances MSTSP21 and MSTSP25 were selected. The evolutionary distribution of their solution sets throughout the algorithm's execution is presented.

As illustrated in Fig. 10, the solution distribution exhibits significant variations across different search stages of the algorithm. During the initial stage, a larger niche radius was configured, resulting in fewer paths sharing identical objective function values, as numerous similar paths were eliminated. This strategy effectively promoted extensive exploration of the decision space, manifested by a dispersed spatial distribution of population individuals. Upon transitioning to the intermediate stage, with a progressive reduction in niche radius and an intensification of the local search operator, the population gradually converged. The algorithm demonstrated an increased tendency to aggregate towards both the global optimum and local optima. In the final stage, the introduction of a population restart mechanism enabled the algorithm to effectively escape the confinement of local optima, thereby facilitating the discovery of additional global optimal solutions.

Fig. 11 illustrates the convergence process of the F_β and DI metrics for the MSLS-MA algorithm on the MSTSP21 benchmark problem. As depicted, during the initial evolutionary stage, the algorithm exhibits high population diversity, attributed to the effective maintenance of solution space diversity facilitated by the introduced niche strategy. As shown in Fig. 10, this stage features a significant number of individuals distant from known optimal solutions, resulting in a relatively slower initial convergence rate for MSLS-MA. As the evolution progresses, the niche radius is progressively reduced, and lower-quality solutions are selectively pruned, leading to a significant enhancement in both the algorithm's convergence rate and the quality of the solution distribution. Concurrently, the application of the local search operator enables the algorithm, upon discovering one optimal solution, to efficiently locate additional functionally equivalent optimal solutions characterized by identical objective function values but distinct paths. Notably, during the later evolutionary stages, fluctuations in the F_β metric arise approximately every 20 iterations. This behavior occurs due to the activation of the population restart mechanism when solution quality stagnates for a prolonged period, facilitating the exploration of potential optimal solutions.

Overall, the two-stage evolutionary framework proposed in this study effectively balances convergence and diversity. Specifically, during the exploration stage, the niche mechanism supplies the algorithm with critical information

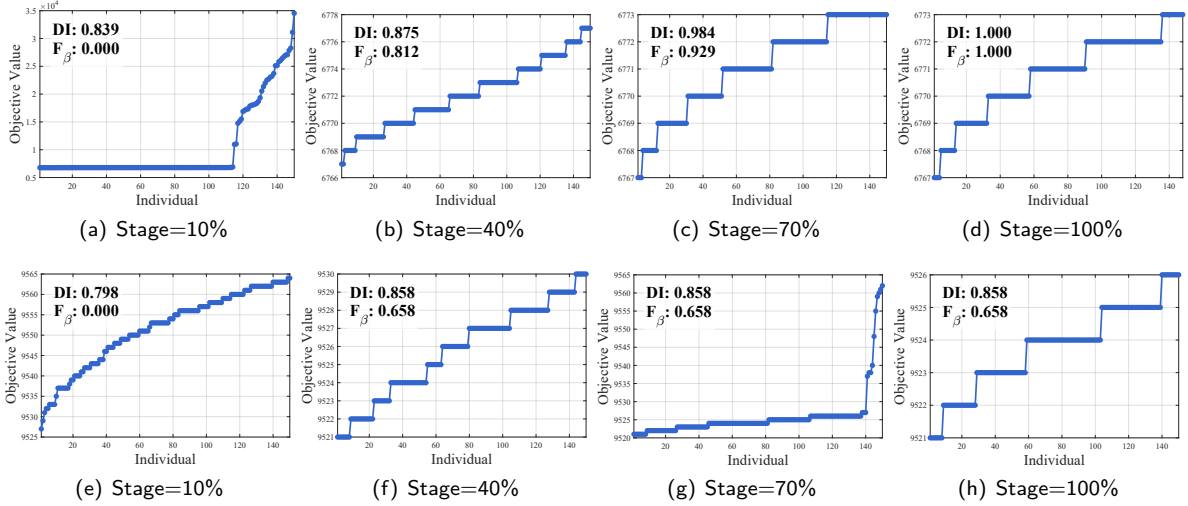


Figure 10: Distribution of solutions in the MSTSP21 (the first row) and the MSTSP25 (the last row) by MSLS-MA on different stages.

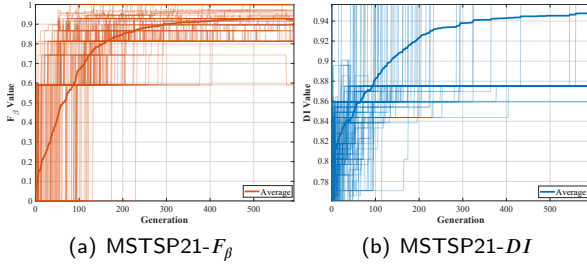


Figure 11: Convergence curves of F_β and DI on MSTSP21 with 48 decision variables obtained by MSLS-MA.

about the decision space, effectively guiding the population towards convergence to the global optimal solution set. In the exploitation stage, the local search operator leverages the information acquired during exploration to drive a refined search within the population. Benefiting from this synergistic two-stage mechanism, the MSLS-MA algorithm successfully discovers all functionally equivalent optimal paths for the MSTSP. Furthermore, it achieves solutions with favorable convergence quality when applied to generalized complex TSP instances.

6. Conclusion

This paper proposes a two-stage memetic algorithm for solving the MMTSP, which effectively integrates exploration and exploitation stages to simultaneously locate multiple optimal solutions. The exploration stage maintains population diversity during the search process, while the exploitation stage ensures convergence capability. To further enhance algorithmic performance, three advanced mechanisms are designed: 1) Adaptive niche mechanism: Dynamically adjusts niche radii to allocate computational resources, achieving more effective exploration-exploitation balance;

- 2) Local search operator: Applies perturbations to converged individuals to mitigate premature convergence risks;
- 3) Population restart mechanism: Provides supplementary guidance and prevents unnecessary deep exploitation during local search.

To validate the algorithm's efficiency and efficacy, MSLS-MA was comparatively evaluated against two existing discrete MMTSP optimizers and four enhanced multimodal evolutionary algorithms. MSLS-MA significantly outperformed all six baselines in both key metrics (F_β and DI). Runtime analysis reveals MSLS-MA's computational complexity exhibits minimal sensitivity to increasing city scales, with average execution time lower than all competitors. Additional experiments on TSPLIB instances further confirm MSLS-MA's superiority in obtaining more optimal and stable solutions even for unimodal TSP variants.

Future research directions include: 1) Scalability enhancement: MSLS-MA's performance on large-scale instances requires improvement. Establishing large-scale MMTSP benchmarks is crucial for effective algorithm evaluation in complex scenarios; 2) Metric refinement: Current metrics primarily focus on solution quality while inadequately capturing solution set characteristics (e.g., inter-solution dissimilarity, feasible solution diversity). Comprehensive evaluation frameworks are urgently needed; 3) Multimodal mechanism optimization: Further investigation into adapting diversity-preservation strategies and population architectures could enhance efficiency for discrete multimodal optimization.

CRedit authorship contribution statement

Lida Zhang: Writing review & editing, Writing original draft, Methodology. **Yan Zheng:** Writing original draft. **Wenhua Li:** Writing review & editing, Supervision, Methodology, Conceptualization. **Rui Wang:** Resources &

Methodology. **Tao Zhang:** Writing review & editing. **Ling Wang:** Writing review & editing.

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